EXPLORATORY DATA ANALYSIS AND VISUALIZATION OF STUDENT ACADEMIC SUCCESS

Case Study Project Report

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Abstract

This project explores a real-world student academic dataset from the UCI Machine Learning Repository. The objective is to use exploratory data analysis and statistical visualization to identify patterns in student characteristics and academic outcomes, and to develop a Multilayer Perceptron (MLP) classifier. The workflow includes dataset inspection, preprocessing, multiple visualizations, data splitting, model selection, and test-set evaluation. Insert all numerical results and findings from the executed notebook; do not report unverified results.

1. Introduction and Objectives

Educational institutions can use data analysis to better understand patterns associated with student progression. This project applies EDA and visualization to a public dataset and demonstrates a reproducible classification workflow. Objectives: inspect data quality; explore distributions and relationships; create and interpret at least five visualizations; train and evaluate an MLP; and document limitations and future improvements.

2. Dataset Description

Dataset: Predict Students' Dropout and Academic Success. Source: UCI Machine Learning Repository. It contains demographic, academic, and socioeconomic attributes; the target field represents Dropout, Enrolled, or Graduate. Insert the exact record and field counts printed by the notebook. Reference: https://doi.org/10.24432/C5MC89

3. EDA and Preprocessing

The notebook checks data types, missing values, duplicate records, descriptive statistics, and target-class frequencies. Exact duplicate rows are removed. Numeric values are imputed with the median and standardized; categorical values are imputed with the most frequent category and one-hot encoded. The preprocessing pipeline is fitted on training data only to reduce data leakage.

4. Data Visualization

Figure 1: target distribution—compare class counts and class imbalance. Figure 2: numeric histograms—describe spread and skewness. Figure 3: correlation heatmap—identify numeric associations without claiming causation. Figure 4: boxplots—inspect spread and potential outliers. Figure 5: numeric feature by outcome—compare distributions across outcomes. Insert the actual plots and specific observations from the executed notebook.

5. Seven Stages of Visualization

1. Acquire: download the dataset. 2. Parse: inspect columns and data types. 3. Filter: remove duplicates and select features. 4. Mine: calculate statistics and correlations. 5. Represent: choose appropriate charts. 6. Refine: add labels, titles, and readable scales. 7. Interact: inspect notebook tables and rerun cells to examine features.

6. Deep Learning Model

A Multilayer Perceptron classifier is used for tabular classification. Two hidden-layer configurations are compared using validation accuracy, with early stopping. Insert the selected configuration, iteration count, and validation score from the actual run.

7. Evaluation and Interpretation

The held-out test set is evaluated using accuracy, precision, recall, F1-score, a confusion matrix, training-loss curve, one-vs-rest ROC curves and AUC values, and per-class metrics. Insert actual outputs and explain which classes are easiest or hardest to predict.

8. Conclusion and Future Scope

Complete after running the notebook. Summarize observed data patterns, data-quality issues, model performance, and limitations. Future work can compare tree-based models, tune hyperparameters, validate on newer cohorts, and assess fairness. Predictions should support—not replace—academic counselling.

9. References

Realinho et al. (2022), UCI Machine Learning Repository, https://doi.org/10.24432/C5MC89. Pedregosa et al. (2011), Scikit-learn: Machine learning in Python, Journal of Machine Learning Research, 12, 2825–2830. Hunter (2007), Matplotlib: A 2D graphics environment, Computing in Science & Engineering, 9(3), 90–95. Waskom (2021), Seaborn: Statistical data visualization, Journal of Open Source Software, 6(60), 3021. McKinney (2010), Data structures for statistical computing in Python, Proceedings of the 9th Python in Science Conference, 56–61.

Appendix A: Source Code

Complete source code is provided in Student_Performance_EDA_Project.ipynb. Run all notebook cells in Google Colab and include the executed notebook in the GitHub repository.